

Data Structure Lab Assignment (CS 2172)

Optional Assignment: Stack applications

Time: As per your suitability.

1. Develop a subroutine that evaluates a postfix expression using Stack.

Phase-1: Provide support only for single digit operand

Phase-2: Provide the support for multi-digits operand

Phase-3: Do necessary changes to provide support to evaluate a prefix expression.

2. Develop a subroutine that converts an infix expression to postfix expression using Stack.

Phase-1: Only support following operators: $+$, $-$, $*$, $/$

Phase-2: Also support additional operators: $($, $)$, $^$

Phase-3: Provide the support for multi-digits operand

Phase-4: Do necessary changes to provide support to convert infix expression to prefix expression.

Note: Use your existing array based integer stack APIs to develop these subroutines.